

Technology Stack

Frontend

- React.js
- HTML5
- CSS3
- JavaScript (ES6+)

Backend

- Supabase
 - Authentication
 - PostgreSQL Database
 - Storage for Images and Videos
 - Row Level Security (RLS)

AI & Processing

- JavaScript-based algorithms for evidence processing
- OCR for number plate and text extraction
- AI APIs for complaint generation and incident summarization
- Speech-to-Text processing for voice descriptions

Hosting & Deployment

- Vercel (Frontend Hosting)
- GitHub (Version Control and Collaboration)

Database

- Supabase PostgreSQL

High-Level Architecture

User → React Frontend → Supabase Backend

1. User uploads photos/videos/audio.
2. Files are stored in Supabase Storage.
3. Metadata is saved in PostgreSQL.
4. OCR and AI modules process evidence.
5. Complaint draft is generated.
6. User reviews and submits report.
7. Report and evidence are stored securely.

8. Status tracking is available through the dashboard.

Core Modules

1. Authentication Module

- User Registration
- Login
- Password Recovery

2. Incident Reporting Module

- Create Incident
- Incident Description
- Draft Management

3. Evidence Management Module

- Upload Photos
- Upload Videos
- Evidence Preview
- Evidence Storage

4. AI Processing Module

- OCR Number Plate Detection
- Speech-to-Text Conversion
- Evidence Analysis
- Complaint Generation

5. Complaint Management Module

- Auto-Generated Complaint
- Edit Complaint
- Export PDF

6. Dashboard Module

- View Submitted Reports
- Track Complaint Status
- Manage Profile

Development Workflow

1. Design UI in React.
2. Configure Supabase Authentication.

3. Create Database Tables.
4. Integrate Evidence Upload Feature.
5. Implement OCR and AI Services.
6. Generate Complaint Drafts.
7. Test Complete Workflow.
8. Push Code to GitHub.
9. Deploy Application on Vercel.
10. Conduct User Testing and Improvements.